

Java Program: NumberFormatException Handling

1. Aim

To write a Java program that demonstrates a NumberFormatException when an invalid string is converted into an integer and handles the exception using a try-catch block.

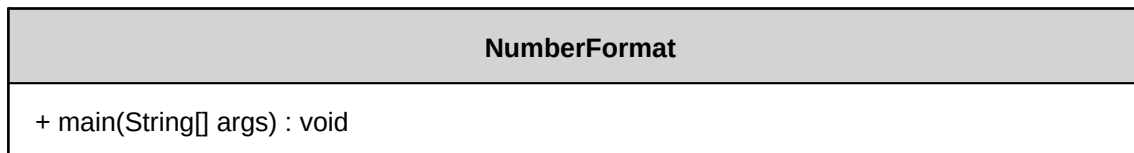
2. Description

A NumberFormatException occurs when a string does not contain a valid numeric value but the program tries to convert it into a number. In this program, the string "abc" is passed to **Integer.parseInt()**. Since "abc" cannot be converted into an integer, Java throws a NumberFormatException. The catch block catches the exception and displays an appropriate error message.

3. Algorithm

1. Start the program.
2. Enter the main method.
3. Create a String variable named invalidNumber and assign the value "abc".
4. Use Integer.parseInt() to convert the string into an integer.
5. Since the string is not a valid number, a NumberFormatException is generated.
6. Catch the exception using a catch block.
7. Display a message indicating that the numeric format is invalid.
8. Stop the program.

4. UML Diagram



5. Source Code with Comments

```
public class NumberFormat {  
    public static void main(String[] args) {  
        try {  
            // Store a string that is not a valid integer  
            String invalidNumber = "abc";  
  
            // Attempt to convert the string into an integer  
            // This statement throws NumberFormatException  
            int number = Integer.parseInt(invalidNumber);  
  
        } catch (NumberFormatException e) {  
            // Handle the invalid numeric conversion  
            System.err.println(  
                "Caught NumberFormatException: "  
                + "Invalid format for numeric conversion."  
            );  
        }  
    }  
}
```

6. Compilation and Execution Commands

```
javac NumberFormat.java  
java NumberFormat
```

The first command compiles the Java source file. The second command executes the compiled program.

7. Output Screenshot

NumberFormatException - Console Output

```
Caught NumberFormatException: Invalid format for numeric conversion.
```

8. Result

The Java program was successfully executed. It demonstrated a `NumberFormatException` when the non-numeric string "abc" was converted to an integer, and the exception was successfully handled using a try-catch block.